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HOTEL UDHAYAM INTERNATIONAL

Hotel Management System

Complete Digital Transformation & Revenue Protection

Phase 1	Phase 2	Phase 3
Core Software System	Management Reports	PIR Sensor Integration

Prepared By: Development Team

Date: February 2026

Version: 1.0

Executive Summary

Project Overview

This proposal outlines a comprehensive digital transformation for Hotel Udhayam International in three phases:

- **Phase 1:** Build complete hotel management software from scratch (Front Desk, Restaurant, Billing, Expenses, Receipts, Inventory, Staff Management)
- **Phase 2:** Digitize management reports (A, B, J formats) with instant WhatsApp sharing to management
- **Phase 3:** Implement PIR sensors for real-time room occupancy detection and revenue leakage prevention

Current Challenges

Current State

- Manual paper-based reports
- Time-consuming data entry
- Delayed reporting to management
- No audit trail
- Revenue leakage undetected

After Implementation

- Automatic digital reports
- One-click data entry
- Instant WhatsApp delivery
- Complete audit trail
- Real-time leakage detection

Expected Benefits

2-3 hrs
Daily Time Saved

100%
Audit Coverage

Investment Overview

Phase	Components	Duration	Investment
Phase 1	Complete Hotel Management Software	6-8 weeks	Software Development
Phase 2	Management Reports (A, B, J) + WhatsApp	2-3 weeks	Software Only
Phase 3	PIR Sensors (50 rooms) + Integration	2-4 weeks	Rs 75,000 (Rs 1,500/room)

Phase 1: Core Hotel Management Software

1 Complete Software System from Scratch

Build a comprehensive hotel management system covering all daily operations with built-in controls and audit trail.

System Modules

Core System Modules

- FD **Front Desk** Check-in/Check-out
- RS **Reservations** Booking Management
- BL **Billing** Invoices & Payments
- RT **Restaurant** KOT & Orders
- EX **Expenses** Daily Expenses
- RC **Receipts** Daily Receipts
- IN **Inventory** Stock Management
- ST **Staff** User Management

Key Features

Module	Features	Built-in Controls
Front Desk	Check-in, Check-out, Room Status, Guest ID capture, Complimentary meals	Photo verification, Advance enforcement, Audit logging
Restaurant	KOT creation, Bill generation, Room posting, Retail sales	KOT-to-Bill tracking, Cancel authorization with PIN
Billing	Invoice generation, Payment collection, GST calculation	Receipt auto-numbering, Payment mode tracking
Expenses	Daily expense entry, Category management, Approval workflow	User-defined categories, Amount limits
Receipts	Daily receipts entry, Source tracking, Bank deposits	Auto-reconciliation, Variance alerts
Inventory	Stock entry, Consumption tracking, Purchase orders	Stock vs Sales matching, Variance alerts
Staff	User accounts, Role management, Shift assignment	Activity logging, Performance tracking

Phase 1: Module Details

1. Front Desk Module

Check-in Process

Room Selection
Room 205 - Deluxe

Guest Name
John Davidson

ID Number
XXXX-XXXX-1234

Guest ID Photo
[Capture ID]

Guest Face Photo
[Capture Face]

Complimentary Meals
☑ Breakfast: 2 pax ☑ Dinner: 2 pax

Advance Payment Required
Room Rate: Rs 2,500/night | Minimum: Rs 2,500

2. Restaurant Module

Room Orders Tab

Post to Room Account

Room: 205 | Guest: John Davidson

Items: Biryani x2, Coffee x2
Total: Rs 1,000 + GST = Rs 1,050

Post to Room

Retail Sales Tab

Cash/Walk-in Sales

KOT: K-015 | Bill: B-015

Amount: Rs 850
Payment: Cash / Card / UPI

Record Sale

3. Expenses & Receipts Module

Daily Expenses

Particulars	Category	Amount
Vegetables	Groceries	Rs 2,500
Staff lunch	Staff	Rs 1,200
Electricity	Utilities	Rs 5,000
Total		**Rs 8,700**

Daily Receipts

Particulars	Category	Amount
From F/Office	Cash	Rs 50,000
Room Collections	Rooms	Rs 85,500
Restaurant	F&B	Rs 15,500
Total		**Rs 1,51,000**

¡ **User-Defined Categories:** Hotel staff can add/remove expense and receipt categories from Settings. No coding required.

Phase 1: Built-in Audit & Controls

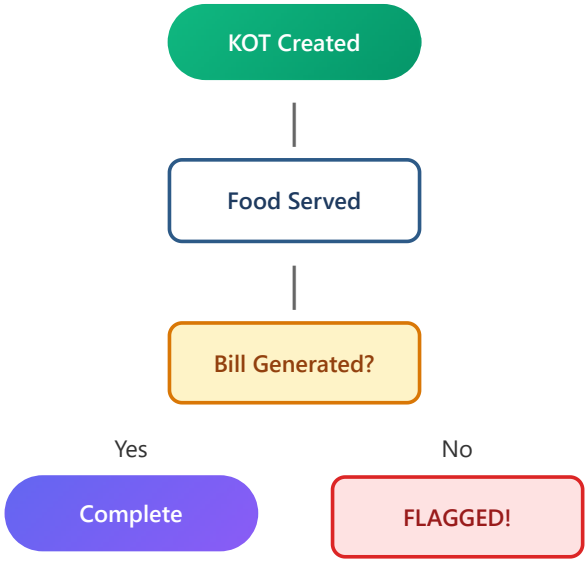
Every Action Leaves a Trail

! **Key Principle:** All actions are logged with user ID, timestamp, and details. Management can review any transaction at any time.

1. Check-in Controls

Control	How It Works	Prevents
Photo Capture	Guest ID and face photo mandatory at check-in	Fake bookings, identity fraud
Advance Enforcement	Minimum 1-night payment required	Walk-away guests, revenue loss
Manager Override	PIN required to waive advance	Unauthorized exceptions
Short Stay Alert	Flag stays less than 4 hours	Unauthorized short-term rentals

2. Restaurant Controls



KOT Cancellation requires:

- Reason selection (Customer cancelled, Wrong order, Quality issue)
- Manager PIN authorization
- Logged in daily report

3. Activity Logging

Activity Log

Time	User	Action	Details
10:30	John	CHECK_IN	Room 205, Guest: Davidson, Advance: Rs 5,000
11:15	Ravi	KOT_CREATE	KOT K-015, Room 205, Amount: Rs 1,050
11:45	Ravi	BILL_GENERATE	Bill B-015, KOT K-015, Paid: Cash
12:00	Ravi	KOT_CANCEL	KOT K-016, Reason: Customer cancelled, PIN: Manager

```
14:30    John    EXPENSE_ADD         Vegetables, Rs 2,500, Category: Groceries
```

Phase 2: Management Reports & WhatsApp

2 Digital Reports with WhatsApp Sharing

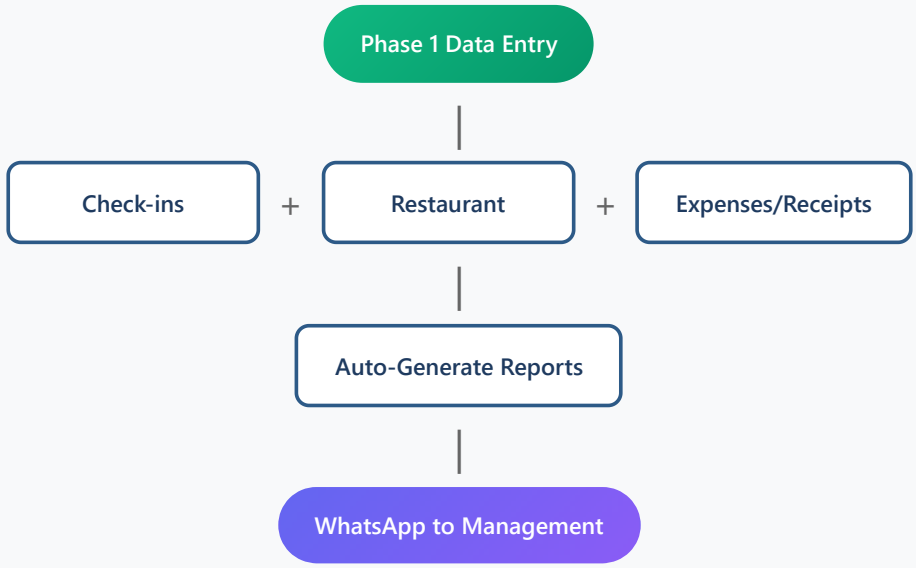
Transform paper-based daily reports (A, B, J formats) into digital formats with instant WhatsApp sharing to management.

Report Formats

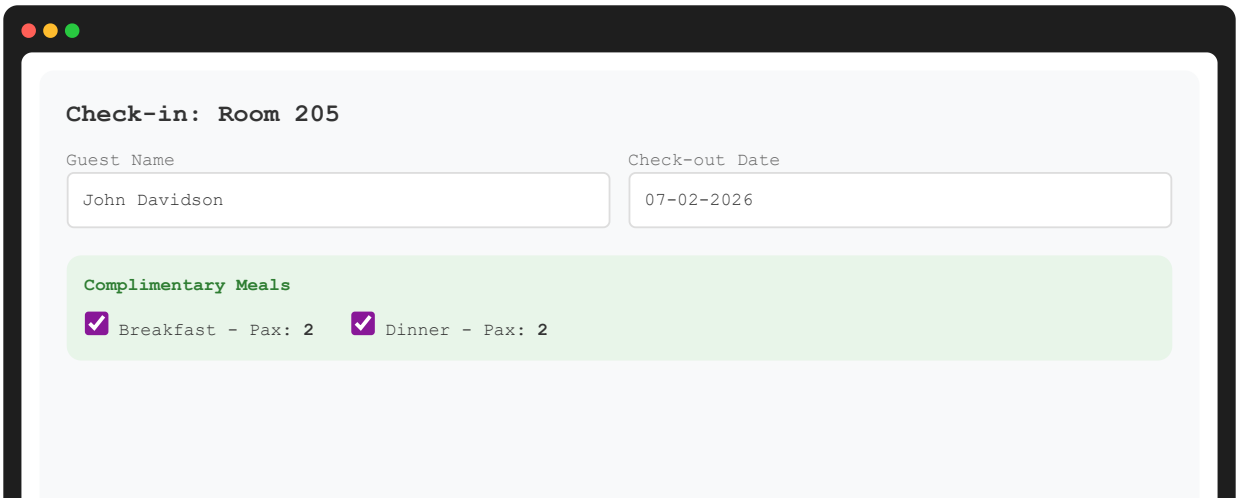
Three Management Report Formats

A	B	J
Format-A	**Format-B**	**Format-J**
Shift Cashier Report	Daily Accounts	Restaurant Sales
Guest-wise payments	Receipts & Payments	KOT & Bill tracking

Data Flow



Complimentary Meals Notification



Complete Check-in

Consolidated Daily Notification (5:00 AM)

DAILY COMPLIMENTARY MEALS

Date: 06-02-2026 | Sent at 5:00 AM

TODAY'S BREAKFAST (10 pax):
- Room 101: 2 pax (Sharma)
- Room 205: 2 pax (Davidson)
- Room 301: 4 pax (Singh family)
- Room 405: 2 pax (Kumar)

TODAY'S DINNER (6 pax):
- Room 205: 2 pax (Davidson)
- Room 301: 4 pax (Singh family)

Auto-sent to Restaurant

¡ **Benefit:** Restaurant receives consolidated daily notification at 5:00 AM with all complimentary meals for the day. One message instead of multiple per-guest notifications.

2. Expenses & Receipts Entry

Expenses Page

Daily Expenses

Particulars	Amount
Vegetables	Rs 2,500
Staff lunch	Rs 1,200
Electricity	Rs 5,000
Total	**Rs 8,700**

Receipts Page

Daily Receipts

Particulars	Amount
From F/Office	Rs 50,000
Room Collections	Rs 85,500
Restaurant	Rs 15,500
Total	**Rs 1,51,000**

Categories are user-configurable - Add/remove expense and receipt categories in Settings.

Phase 2: Report Format Previews

Format-A: Shift Cashier Report

Uses existing Shift Handover page layout with Financial Summary, Operations Summary, and Payment breakdown.

John Doe - Morning Shift
6:00 AM - 2:00 PM | 05-02-2026

7h 45m Duration | **42** Transactions

Rs **Financial Summary**

Opening Balance	Rs 50,000
Room Charges	Rs 85,500
Advance Payments	Rs 24,000
Restaurant & Services	Rs 15,500
Closing Balance	**Rs 92,500**

Rs 45K	Rs 52K	Rs 18K	Rs 10K
Cash	Card	UPI/GPay	Bank

8	5	3
Check-ins	Check-outs	Pending

OB + ADV + RST + CAB + M.HALL + BALANCE = TOTAL - HR - SBI - GPAY - CC - RF = CB

OB	ADV	RST	CAB	M.HALL	BAL	=TOTAL	HR	SBI	GPAY	CC	RF	=CB
50K	16K	2.5K	1.5K	5K	8K	83K	5K	25K	8K	12K	2K	31K

Format-B: Daily Accounts

HOTEL UDHAYAM INTERNATIONAL
DAILY ACCOUNTS ADMINISTRATIONS B-FORMAT
DATE: 05-02-2026 TIME: 7AM to 10PM

Sl.No	Particulars	Receipts	Payments
	Opening Balance Rs. 50,000		
1	Received from F/Office	50,000	
2	Room Collections	85,500	
3	Restaurant Sales	15,500	
4	Vegetables		2,500
5	Staff lunch		1,200
6	SBI Deposit		50,000
	TOTALS	**1,51,000**	**53,700**
	Closing Balance	**Rs 1,47,300**	

Format-J: Restaurant Retail Sales

HOTEL UDHAYAM INTERNATIONAL
FORMAT-J BELL RESTAURANT
DATE: 05-02-2026 TIME: 7:00AM TO 10:30PM

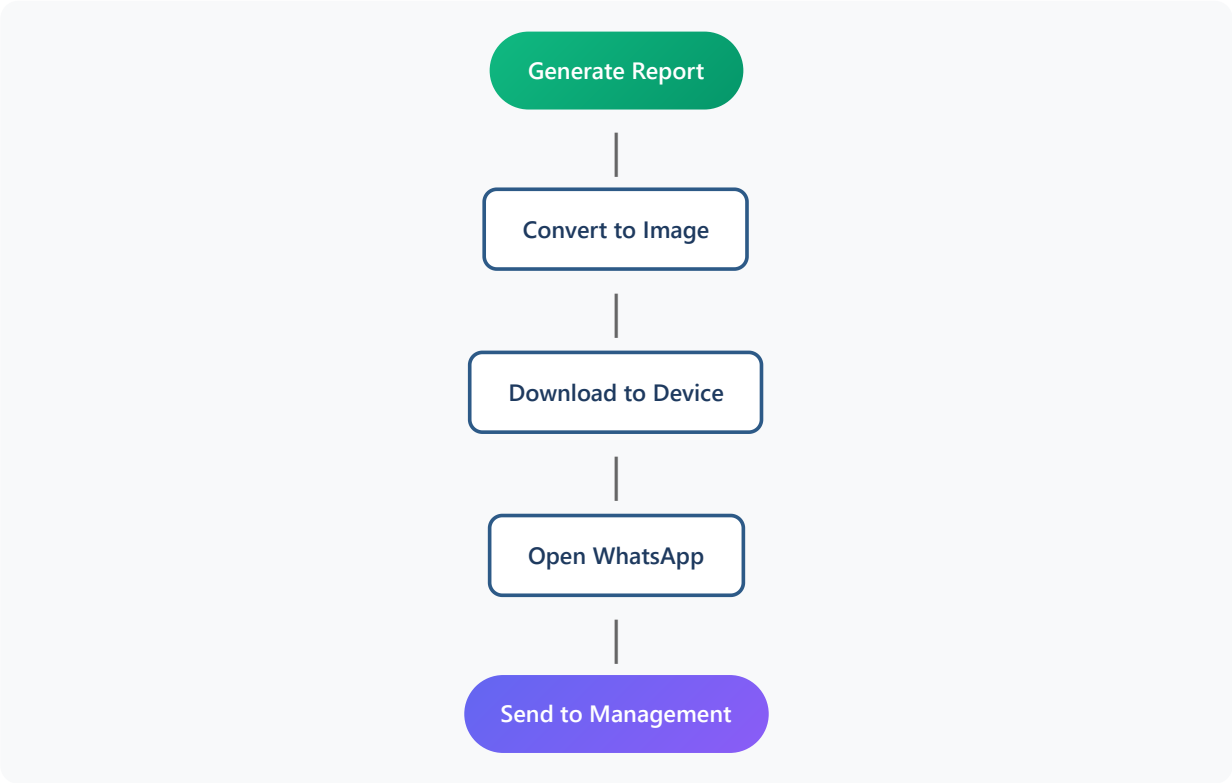
S.No	KOT	Bill	Amount	Room	S.No	KOT	Bill	Amount	Room
1	K001	B001	850	101	6	K006	B006	1,100	205
2	K002	B002	450	CASH	7	K007	B007	780	CASH

3	K003	B003	1,200	205	8	K008	B008	650	301

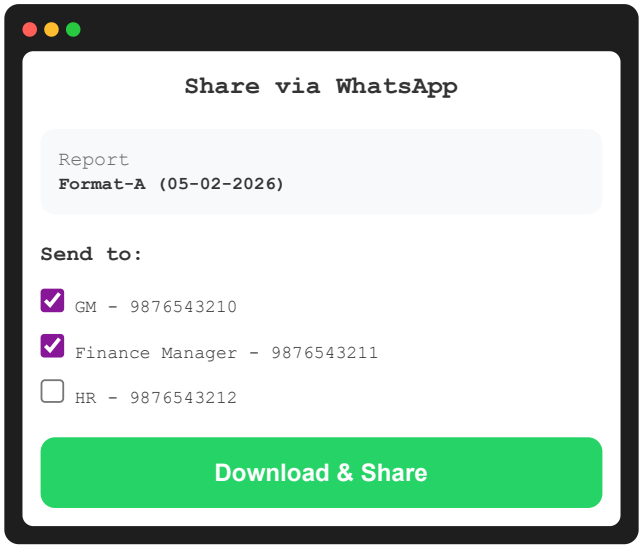
Complimentary: Breakfast - 10 pax | Dinner - 6 pax

Phase 2: WhatsApp Integration

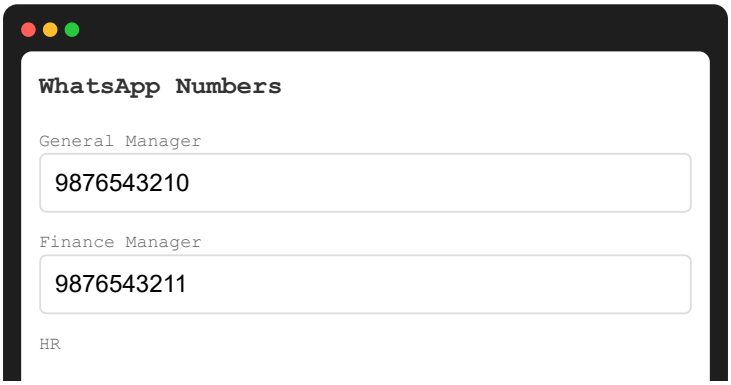
How It Works



Share Modal



Settings Configuration



9876543212

Restaurant

9876543213

Save Numbers

Phase 3: PIR Sensor Integration

3 Real-Time Room Occupancy Detection

Implement PIR sensors in all rooms to detect unauthorized occupancy and prevent revenue leakage through real-time monitoring.

Revenue Leakage Points

! **Problem:** Hotels typically lose 5-15% of revenue through unauthorized transactions that go undetected.

Common Leakage Scenarios

Unauthorized Check-ins
Staff allows short stays, collects cash, no record in system

Pocket Collection
Payment collected from guest but not entered in system

Restaurant Fraud
Orders served but bill not generated, cash pocketed

Inventory Theft
Stock consumed but not accounted in sales

Our Solution

Control	What It Catches	How It Works
PIR Sensors	Unauthorized room occupancy	Detects motion in "Available" rooms
Photo Capture	Fake bookings	Guest ID & face required at check-in
Advance Enforcement	Walk-away guests	Minimum payment required, override needs PIN
KOT Tracking	Unbilled orders	Every KOT must have bill or authorized cancel
Stock Reconciliation	Inventory theft	Match consumption with sales
Activity Logs	Staff misbehavior	All actions logged with timestamp

Phase 3: How PIR Detection Works

How PIR Detects Unauthorized Occupancy

[ROOM 205]

Motion Detected!

System Status	PIR Sensor	Result
Available	No Motion	OK
Available	**MOTION**	**ALERT!**
Occupied	Motion	OK
Maintenance	**MOTION**	**ALERT!**

Real-Time Monitoring Dashboard

Room Occupancy Monitor

Room	System Status	PIR Sensor	Status
101	Occupied	Motion	OK
102	Available	No Motion	OK
205	**Available**	**MOTION!**	**ALERT**
301	Occupied	Motion	OK

1 Room with suspicious activity detected!

Alert Notification

OCCUPANCY ALERT

Room: 205
System Status: Available
PIR Sensor: Motion Detected
Time: 2:30 PM

Possible unauthorized occupancy!
Please verify immediately.

Hardware Cost

Rs 1,500	Rs 75,000	1-2 Months
Per Room (Hardware)	50 Rooms Total	ROI Recovery

Phase 3: Daily Audit Reports & ROI

Daily Summary (Auto WhatsApp at 10 PM)

HOTEL UDHAYAM DAILY SUMMARY

Date: 05-02-2026

ROOMS:
Check-ins: 12 | Check-outs: 8
Occupancy: 75%
Revenue: Rs 1,25,000

RESTAURANT:
Orders: 45 | Cancelled: 3
Revenue: Rs 28,500

CASH FLOW:
Opening: Rs 50,000
Collections: Rs 1,53,500
Expenses: Rs 28,000
Closing: Rs 1,75,500

ALERTS (3):
1. Short stay Room 205 (4 hrs)
2. KOT K-003 unbilled
3. Chicken variance: -2kg

Return on Investment

Expected Monthly Savings

Rs 20-50K Room Leakage Prevention	**Rs 10-30K** Restaurant Leakage Prevention
Rs 5-15K Inventory Theft Prevention	**Rs 50-1L** Total Monthly Savings

Rs 75,000 PIR Investment (50 rooms)	**2-3 Months** Investment Recovery	**24/7** Monitoring

¡ **Key Benefit:** The PIR sensor investment of Rs 75,000 (Rs 1,500/room) can be recovered in just 2-3 months through prevented revenue leakage. After that, all savings go directly to profit.

Implementation Timeline

Phase 1
Core Hotel Management Software

- Front Desk (Check-in/Check-out)
- Reservations & Billing
- Restaurant (Room orders & Retail)
- Expenses & Receipts entry
- Inventory & Staff management
- Built-in audit controls

6-8W

Phase 2
Management Reports & WhatsApp

- Format-A (Shift Cashier Report)
- Format-B (Daily Accounts)
- Format-J (Restaurant Sales)
- Complimentary meals notification
- WhatsApp integration

2-3W

Phase 3
PIR Sensor Integration

- PIR sensor procurement
- Room-by-room installation
- WiFi connectivity setup
- Real-time monitoring dashboard
- Alert notifications
- Testing & calibration

2-4W

Total Project Duration: 10-15 Weeks

Phase	Duration	Investment	Dependencies
Phase 1: Core Software	6-8 weeks	Software Development	None
Phase 2: Reports & WhatsApp	2-3 weeks	Software only	Phase 1 complete
Phase 3: PIR Sensors	2-4 weeks	Rs 75,000 (Rs 1,500/room)	Hardware only

Approval & Sign-off

Project Summary

Total Duration	10-15 weeks
Phase 1 (Software)	Complete hotel management system from scratch
Phase 2 (Reports)	A, B, J formats with WhatsApp integration
Phase 3 (Hardware)	Rs 75,000 (Rs 1,500/room x 50 rooms, hardware only)
Expected Monthly Savings	Rs 50,000 - 1,00,000
ROI Period	2-3 months (after Phase 3)

Management Approval Required

☐ **Phase 1: Core Hotel Management Software**
Front Desk, Restaurant, Billing, Expenses, Receipts, Inventory, Staff Management with built-in audit controls

☐ **Phase 2: Management Reports & WhatsApp Integration**
Digitize A, B, J formats with instant WhatsApp sharing to management

☐ **Phase 3: PIR Sensor Integration**
Hardware-based room occupancy detection (Rs 75,000 for 50 rooms, hardware only)

General Manager	Owner / Director
Name: _____	Name: _____
Date: _____	Date: _____

Hotel Udhayam International

T.B. Road, Tiruchendur

System Enhancement Proposal - Version 1.0

February 2026